

CHEM 101 Topic 9: Stoichiometry Calculations — Problem Set

Instructions: Complete the following problems using the analysis method practiced in your guided assignment. Show all your work, including molar mass calculations, mole ratios, and units at each step.

Section 1: Mole-Mole Calculations

Problem 1: Using the balanced equation $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, how many moles of NH_3 are produced from 4.25 mol of N_2 ?

Problem 2: Using the same equation ($\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$), how many moles of H_2 are required to produce 6.00 mol of NH_3 ?

Problem 3: For the reaction $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$, how many moles of O_2 are needed to react completely with 2.50 mol of Al ?

Section 2: Mole-Mass Calculations

Problem 4: Using $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, how many grams of NH_3 are produced from 10.0 g of N_2 ? (Molar masses: $\text{N}_2 = 28.02 \text{ g/mol}$, $\text{NH}_3 = 17.03 \text{ g/mol}$)

Problem 5: How many grams of Fe are needed to produce 40.0 g of Fe_2O_3 in the reaction $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$? (Molar masses: $\text{Fe} = 55.85 \text{ g/mol}$, $\text{Fe}_2\text{O}_3 = 159.69 \text{ g/mol}$)

Problem 6: In the reaction $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$, if 150 g of C_3H_8 is burned, how many grams of H_2O are produced? (Molar masses: $\text{C}_3\text{H}_8 = 44.11 \text{ g/mol}$, $\text{H}_2\text{O} = 18.02 \text{ g/mol}$)

Section 3: Molar Solutions

Problem 7: How many moles of HCl are in 100 mL of a 0.750 M solution?

Problem 8: What mass of NaCl is needed to prepare 250 mL of a 0.300 M solution? (Molar mass $\text{NaCl} = 58.44 \text{ g/mol}$)

Problem 9: 100 mL of 0.200 M AgNO_3 reacts with NaCl . How many grams of AgCl precipitate will form? ($\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$; Molar mass $\text{AgCl} = 143.32 \text{ g/mol}$)

Section 4: Percent Yield

Problem 10: If a reaction has a theoretical yield of 25.0 g but only 21.2 g is collected, what is the percent yield?

Problem 11: A reaction with a theoretical yield of 50.0 g has a percent yield of 88%. What was the actual yield?

Problem 12: In the reaction $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, 5.00 g of H_2 is reacted. If 40.0 g of H_2O is actually produced, what is the percent yield? (Molar masses: $\text{H}_2 = 2.02 \text{ g/mol}$, $\text{H}_2\text{O} = 18.02 \text{ g/mol}$)